CDAC Advanced Machine Learning — Day 8

Master Technical Summary: Polynomial Regression, Model Complexity & K-Means Clustering

Module: Advanced Machine Learning	**Topics:** Regression & Clustering	**Target:** Exam & Practical Revision	**Author:** abs6187

PART 1: Polynomial Regression & Model Complexity

1.1 Non-Linear Modeling: Standard Simple Linear Regression assumes a strictly monotonic straight-line relationship ($y = mx + c$). However, most real-world data shows curvature: facility operating costs drop before rising again, fertilizer accelerates crop yield up to an optimal peak before toxicity burns the plants, and engine fuel efficiency peaks at mid-range RPMs. Polynomial Regression models these non-linear relationships by creating higher-order polynomial terms while maintaining linear parameter estimation solved via Ordinary Least Squares (OLS).

Mathematical Hypothesis Function:

- **y_hat = $\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + ... + \beta_d x^d$**
- **Matrix Notation:** $\hat{y} = X_{poly}\beta$, where $X_{poly} = [1, x, x^2, ..., x^d]$
- **OLS Loss Function (MSE):** $J(\beta) = (1/n)\sum(y_i - \hat{y}_i)^2 \rightarrow$ Solved via Normal Equation: $\beta = (X^TX)^{-1}X^Ty$

1.2 Polynomial Degree & Curvature Reference Guide

Degree	Mathematical Curve	Bends	Real-World Example	Diagnosis / Risk
Degree 1	Linear: $y = \beta_0 + \beta_1 x$	0 bends	Experience vs Salary	Severe **Underfitting** on curved patterns
Degree 2	Quadratic: $y = \beta_0 + \beta_1 x + \beta_2 x^2$	**1 bend** (U or Bell)	• Temp vs HVAC Cost • Fertilizer vs Crop Yield	**Optimal Fit** for natural single-turn phenomena
Degree 3	Cubic: $y = \beta_0 + ... + \beta_3 x^3$	2 bends (S-curve)	Technology adoption lifecycle	Marginal gains on simple datasets
Degree 4+	Quartic: $y = \beta_0 + ... + \beta_4 x^4$	3+ bends (W-curve)	Complex financial oscillations	High **Overfitting** risk (memorizes noise)

1.3 Scikit-Learn Pipeline Implementation (make_pipeline)

```
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import make_pipeline

# Pipeline chains feature expansion and OLS linear regression
model = make_pipeline(
    PolynomialFeatures(degree=2, include_bias=False), # [x] -> [x, x^2]
    LinearRegression()                                # Fits: y_hat = b0 + b1*x + b2*x^2
)
model.fit(X_train, y_train)

# Extract learned parameters from pipeline step
lr = model.named_steps["linearregression"]
b0, (b1, b2) = lr.intercept_, lr.coef_
```

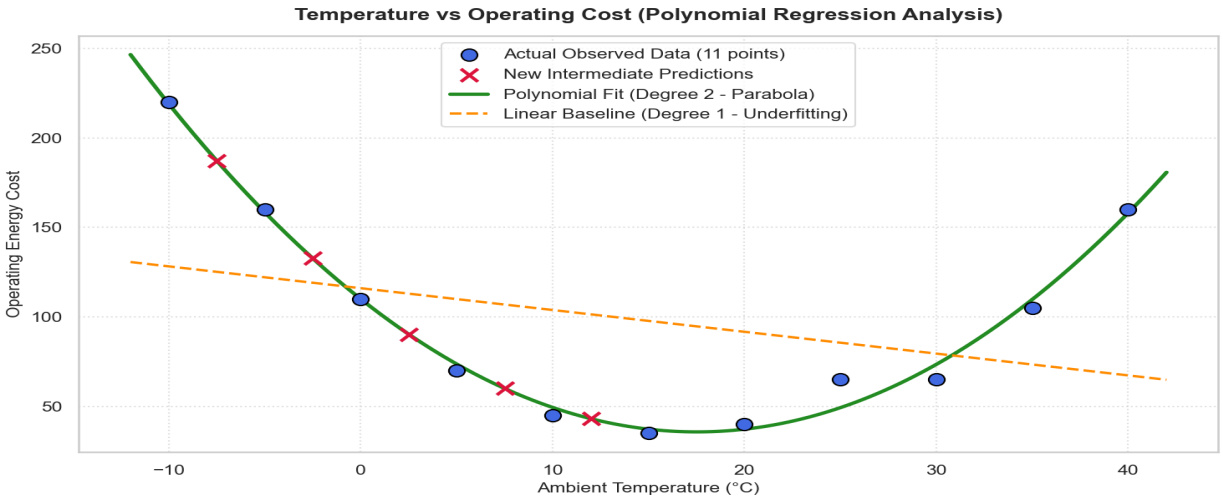
Rule on include_bias=False: By default, **`PolynomialFeatures`** prepends a column of ones. Because **`LinearRegression(fit_intercept=True)`** automatically estimates the intercept term β_0, we set **`include_bias=False`** to prevent duplicate collinear intercept columns.

1.4 Practical Class Problems Walkthrough

Problem 1: Temperature vs Facility Operating Cost (Parabolic U-Curve)

Heating electricity drops as outdoor temperature warms from freezing (-10°C to 0°C), hits minimum in the comfortable room zone (15°C to 20°C), then climbs back up due to air conditioning load (25°C to 40°C).

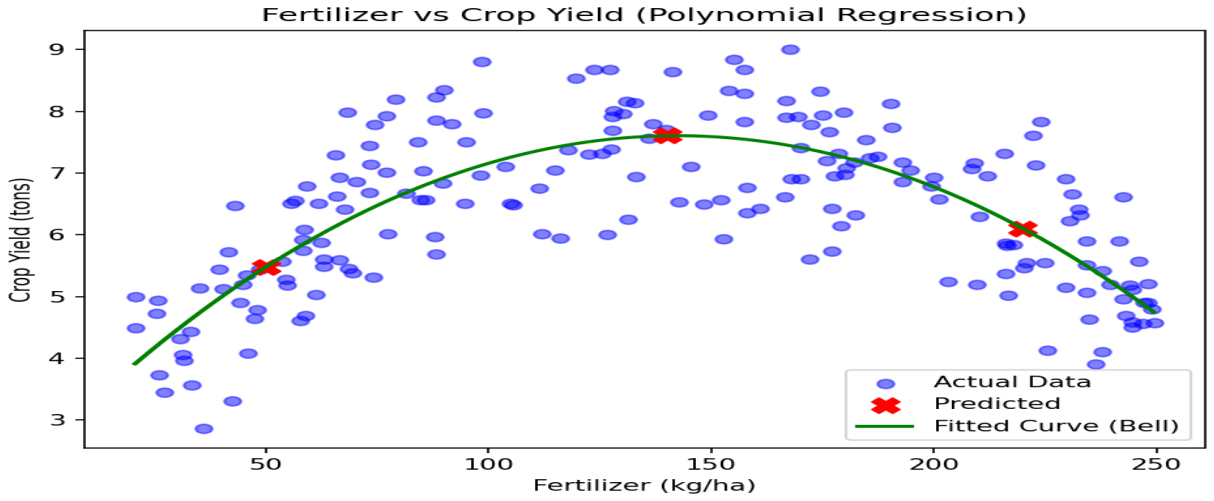
- **Fitted Model: Cost(T) = 109.95 - 8.48·T + 0.24·T^2**
- **Performance:** Degree 1 RMSE: 53.76 (Underfitting) → **Degree 2 RMSE: 5.98 (Optimal Fit)**



Problem 2: Fertilizer Application vs Crop Yield (Bell / Arch Curve)

Agronomic relationship where fertilizer initially accelerates crop yield, reaches an optimal peak yield at ~142.3 kg/ha, beyond which excess chemical fertilizer causes toxicity/burn and reduces crop yield.

- **Generator Formula:** Yield = 2.0 + 0.08·F - 0.00028·F^2 + Noise
- **Fitted Model: Yield = 2.56 + 0.071·F - 0.00025·F^2**
- **Performance:** Degree 1 R^2: 0.0096 (Flat line) → **Degree 2 R^2: 0.6013 (Explains all non-noise variance)**



PART 2: Unsupervised Learning & K-Means Clustering

2.1 Core Algorithm Mechanics (Lloyd's Algorithm): K-Means is a centroid-based, iterative partitioning algorithm that divides an unlabeled dataset of *N* observations into *K* distinct, non-overlapping clusters. The objective is to minimize within-cluster variance (Inertia).

Phase	Mathematical Operation	Core Logic
1. Initialization	Choose K initial centroid vectors: $\mu_1, \mu_2, ..., \mu_K$	Done randomly or strategically via **K-Means++**
2. Assignment	$c^{(i)} := \text{argmin}_j \|\|x^{(i)} - \mu_j\|\|^2$	Assign each sample to its nearest centroid (Euclidean distance)
3. Update	$\mu_j := (1 / \|C_j\|) \Sigma x^{(i)}$ (for i in C_j)	Recompute centroid as the mean vector of all points in cluster
4. Convergence	$\|\|\mu_j^{(t+1)} - \mu_j^{(t)}\|\| < tol$	Repeat steps 2 & 3 until centroids stabilize or max_iter reached

2.2 Why K-Means++ is Crucial

Standard random initialization frequently selects initial centroids close together, causing slow convergence and trapping the algorithm in poor local minima. **K-Means++** solves this: it selects the first centroid randomly, then selects each subsequent centroid with probability proportional to the squared distance $D(x)^2$ from the nearest existing centroid. This ensures initial centroids are spread widely across the data space.

2.3 Why Feature Scaling (StandardScaler) is Mandatory

Because K-Means measures proximity using Euclidean distance (L_2 norm: $d = sqrt(\Sigma(x_j - y_j)^2)$), features with larger raw numeric ranges will completely dominate cluster calculations. For example, if 'Annual Income' ranges from $20,000 to $150,000 and 'Age' ranges from 18 to 70, income will dictate 99.9% of distance computations. Running `StandardScaler()` beforehand guarantees zero mean and unit variance across all dimensions.

2.4 Determining Optimal K (Hyperparameter Tuning)

• **The Elbow Method (Inertia / WCSS):** Plots Inertia ($\Sigma||x - \mu||^2$) against *K*. As *K* increases, Inertia always decreases. Look for the 'elbow' point where the rate of decrease abruptly slows down (point of diminishing returns).
• **Silhouette Analysis:** Evaluates both within-cluster cohesion (*a*) and between-cluster separation (*b*): $s = (b - a) / max(a, b)$. Ranges from -1 to +1. Peak score indicates optimal compactness and separation.
• **Cluster Profiling:** Always invert-transform fitted centroid coordinates back into original units (`scaler.inverse_transform(kmeans.cluster_centers_)`) to give actionable business labels to clusters.

2.5 Key Limitations of K-Means

• **Assumes Convex / Spherical Clusters:** Partitions space with linear Voronoi boundaries. Fails on concentric rings, moons, or arbitrary density manifolds (use DBSCAN instead).
• **Outlier Sensitivity:** Because centroids compute arithmetic means, extreme outliers distort centroid positions.
• **Fixed K Requirement:** You must specify *K* in advance; the algorithm cannot discover cluster count automatically.

PART 3: Evaluation Metrics Quick Reference Table

Metric	Submodule	Mathematical Formula	Key Insight / Target
MAE	Regression	$(1/n) \Sigma \lvert y_i - y_hat_i \rvert$	Interpretable in target units; robust against extreme outliers.
MSE	Regression	$(1/n) \Sigma (y_i - y_hat_i)^2$	Quadratic error penalty; heavily penalizes large errors.
RMSE	Regression	sqrt(MSE)	Brings MSE penalty back to target unit scale. Primary regression metric.
R^2 Score	Regression	$1 - (SS_{res} / SS_{tot})$	Variance explained (0 to 1). 1.0 = perfect prediction; 0 = baseline mean.
Adjusted R^2	Regression	$1 - [(1 - R^2)(n-1)/(n-p-1)]$	Penalizes non-informative features added to multiple regression.
Inertia (WCSS)	Clustering	$\Sigma \Sigma \lVert x - \mu_k \rVert^2$	Within-cluster compactness. Lower is better, used in Elbow plot.
Silhouette Score	Clustering	(b - a) / max(a, b)	Range [-1, +1]. High positive value = dense, well-isolated clusters.

PART 4: Key Exam & Viva Multiple-Choice Questions

Q1: Why is feature scaling (e.g., StandardScaler) essential before applying K-Means clustering?

Correct Answer: B — K-Means relies on Euclidean distance, so unscaled features with larger magnitude ranges will disproportionately dominate cluster assignments.

Explanation: K-Means calculates distance vectors using L2 norm. Unscaled features with large variance dictate centroid placement entirely.

Q2: What problem does the K-Means++ algorithm specifically address?

Correct Answer: C — Choosing initial centroid locations strategically to avoid bad local minima and speed up convergence.

Explanation: K-Means++ picks initial centroids probabilistically based on squared distance from existing centroids, preventing cluster collapse.

Q3: What does a Silhouette Score of +0.85 indicate for a clustered dataset?

Correct Answer: B — Clusters are dense, well-separated, and clearly distinct from one another.

Explanation: The Silhouette score ranges from -1 to +1. Values near +1 represent high within-cluster cohesion and strong between-cluster separation.

Q4: Which scenario is K-Means NOT suitable for?

Correct Answer: C — Clustering concentric circle datasets or arbitrary non-convex density shapes.

Explanation: K-Means uses Voronoi partitioning, creating linear decision boundaries. It cannot separate manifold or ring structures (use DBSCAN).